

Ashwini Kumar

Roll No: 2023ugcm004

B.Tech Engineering and Computational Mechanics

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Education

National Institute of Technology, Jamshedpur

B.Tech in Engineering and Computational Mechanics CGPA: 7.71/10

2023 – 2027

Jamshedpur, India

Delhi Public School, Ranchi

Intermediate (Class XII) Percentage: 93.4%

2020 – 2022

Ranchi, India

MDNGGS Public School, Medininagar

Matriculation (Class X) Percentage: 94.4%

2020

Medininagar, India

Projects

FinSight AI – AI-Powered Investor Intelligence Platform [GitHub](#)

Technologies Used: Python, FastAPI, OpenAI API, Azure PostgreSQL, LangChain, Vector Store, Docker, RAG

- Built an end-to-end financial document intelligence platform to parse annual report PDFs into structured Markdown and extract key financial KPIs (Revenue, Net Income, Cash Flow, Debt).
- Engineered a Retrieval-Augmented Generation (RAG) research pipeline utilizing vector search and OpenAI / Azure OpenAI GPT models to deliver grounded Q&A with source citations.
- Developed an interactive investor dashboard with FastAPI to display company-to-company financial comparisons, document search, and real-time metric analytics.
- Containerized the application using Docker and configured AKS deployment manifests with automatic database seeding and health monitoring.

Multi-Class Email Security & Threat Classifier [GitHub](#) | [Live Demo](#)

Technologies Used: Python, Scikit-Learn, Streamlit, SHAP, TF-IDF, Pandas, Plotly

- Fused 5,000 TF-IDF text n-grams with 20 structured metadata signals (domain reputation, link ratios, urgency flags) to classify emails into 6 categories (Phishing, Spam, Promotion, Social, Updates, Primary).
- Trained Logistic Regression, Random Forest, and Calibrated LinearSVC pipelines using 5-fold cross-validation grid search and balanced class weighting, achieving 99 % macro-F1 performance on benchmark evaluation splits.
- Containerized & Cloud Deployed: Containerized the Streamlit web application using Docker with automated health checks and deployed to cloud hosting for real-time inference and audit report downloads.

FlashKart – Multi-Vendor E-Commerce Platform [GitHub](#) | [Live Demo](#)

Technologies Used: Next.js, PostgreSQL (Neon), Prisma, Stripe, Clerk, Redux, Inngest, ImageKit, Tailwind CSS

- Engineered an AI-powered semantic vector search engine using Google Gemini 768-dim embeddings and cosine distance in PostgreSQL.
- Architected a multi-vendor marketplace with Next.js 15 and Clerk RBAC across Customer, Seller, and Admin portals with automated order splitting across 45+ dynamic routes.
- Integrated a secure Stripe payment and webhook pipeline with server-side price validation, signature verification, and idempotent processing via Inngest.
- Built real-time seller/admin analytics dashboards (Recharts, Prisma) with GMV tracking, inventory controls, and review analytics.

Technical Skills

- **Languages:** Python, C++, C, JavaScript, TypeScript, SQL
- **AI / LLMs:** Machine Learning, LangChain, RAG, Vector Databases, Google Gemini API
- **Frontend:** HTML, CSS, React.js, Next.js, Redux
- **Backend:** Node.js, Express.js, REST APIs
- **Tools & Databases:** Docker, Git, GitHub, Docker, Vercel, PostgreSQL, MongoDB, MySQL
- **Computer Science Fundamentals:** DSA, OOP, Operating Systems, DBMS, Computer Networks

Achievements

- Selected participant, Amazon ML Summer School 2025, among thousands of applicants.
- Solved 1000+ DSA problems across LeetCode, Codeforces, GeeksforGeeks and CodeChef.
- Earned a verified 3-Star profile badge rating on CodeChef.
- Achieved a peak competitive rating of 1540 on Leetcode and 1196 on Codeforces.

Position of Responsibility

- **Co-Founder, CMDC** – Led Machine Learning initiatives in the Computational Mechanics Design Club.
- **NSS Member & Sankalp Volunteer** – Mentored underprivileged students and contributed to social outreach programs.